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PALEOMAGNETIC DATA FROM SARAWAK, MALAYSIAN BORNEO, AND THE LATE MESOZOIC AND CENOZOIC TECTONICS OF SUNDALAND

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Abstract. Paleomagnetic data from 231 samples from 31 sites in rocks of Upper Jurassic to Miocene age in Sarawak (Malaysian Borneo) reveal a trend of increasing counterclockwise (CCW) declination deflection with age. Six sites in Tertiary hypabyssal intrusions show 8° to 52° of CCW deflection. The intrusion deflected 52° CCW was K-Ar dated at 26 m.y. (Upper Oligocene), while one deflected 22° CCW gives a 17 m. y. age (Lower Miocene). Three sites in the Upper Eocene to Miocene(?) Silantek Formation show an average 40° of CCW deflection. Prefolding directions, showing 90° of CCW deflection, are isolated in 4 sites (including two positive fold tests) in Upper Jurassic and Cretaceous rocks of the Bau Limestone and Pedawan Formations. A postfolding, Cenozoic remagnetization with an average of 60° of CCW deflection is found in five Bau Limestone sites. Three sites in the Upper Jurassic Kedadom Formation show an average of 50° of CCW deflection. CCW declination deflections found in Mesozoic and Cenozoic rocks as far as 400 km east and 150 km south of Sarawak, in Kalimantan (Indonesian Borneo), also fit the trend of deflection versus age. On the basis of the regional consistency of declination deflection versus age, along with geologic evidence the data are considered to be evidence of a regional (rather than a local block or distributed shear) rotation. The domain of CCW rotation extends into West Malaysia, suggesting that West Borneo and the Malay Peninsula may have been a stable block during the latest Cretaceous and Cenozoic. West Malaysia and Borneo may have had different histories in the rest of the Mesozoic. The data imply up to 108° CCW rotation of Borneo with respect to stable Eurasia, sometime during the Cretaceous and Cenozoic. Cenozoic rotation may also have occurred between Indochina and Borneo. The sense of rotation shown by the data does not support the "propagating extrusion tectonics" model for Cenozoic Southeast Asia.

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INTRODUCTION

Southeast Asia represents a unique forum in which to study the results of continent-continent collision using paleomagnetism. Major plate tectonic events such as the collision of India with the Eurasian margin about 50 Ma, the change in Pacific plate motion about 45 Ma, and the Miocene impingement of the Australian continent on the southeastern margin of Southeast Asia have all had a profound effect upon the tectonic development of the region.

The extreme tectonic complexity of Southeast Asia, limited knowledge of its geology, and a shortage of geophysical data have combined to make attempts at paleogeographic reconstructions very difficult. The limited amount of paleomagnetic data which exists for Southeast Asia is largely insufficient for use in reconstructions due to undersampling, lack of stability tests, and unreliable age control. A detailed paleomagnetic data base is necessary to provide absolute constraints on crustal motions. When it is complete, it will be possible to describe apparent polar wander paths (APWP) for the larger continental crustal blocks in Southeast Asia and to quantify their relative motions. Paleomagnetic poles derived from the many smaller terranes of Indonesia and the Philippines can also be compared to the APWPs to help explain terrane accretion kinematics in Southeast Asia.

Sundaland is an amalgamation of continental crust which includes most of the Mesozoic and older cratonic regions in Southeast Asia (Figure 1). The exact borders of Sundaland are not yet defined, nor is the extent of its internal deformation. A considerable amount of reconnaissance paleomagnetic data has been presented from Sundaland (pre-1982 described by Haile and Briden [1982], Achache and Courtillot [1985], Maranate and Vella [1986], Untung et al. [1987], and McCabe et al. [1988]). However, most of these data come from Mesozoic rocks in West Malaysia and Thailand. Borneo is central to

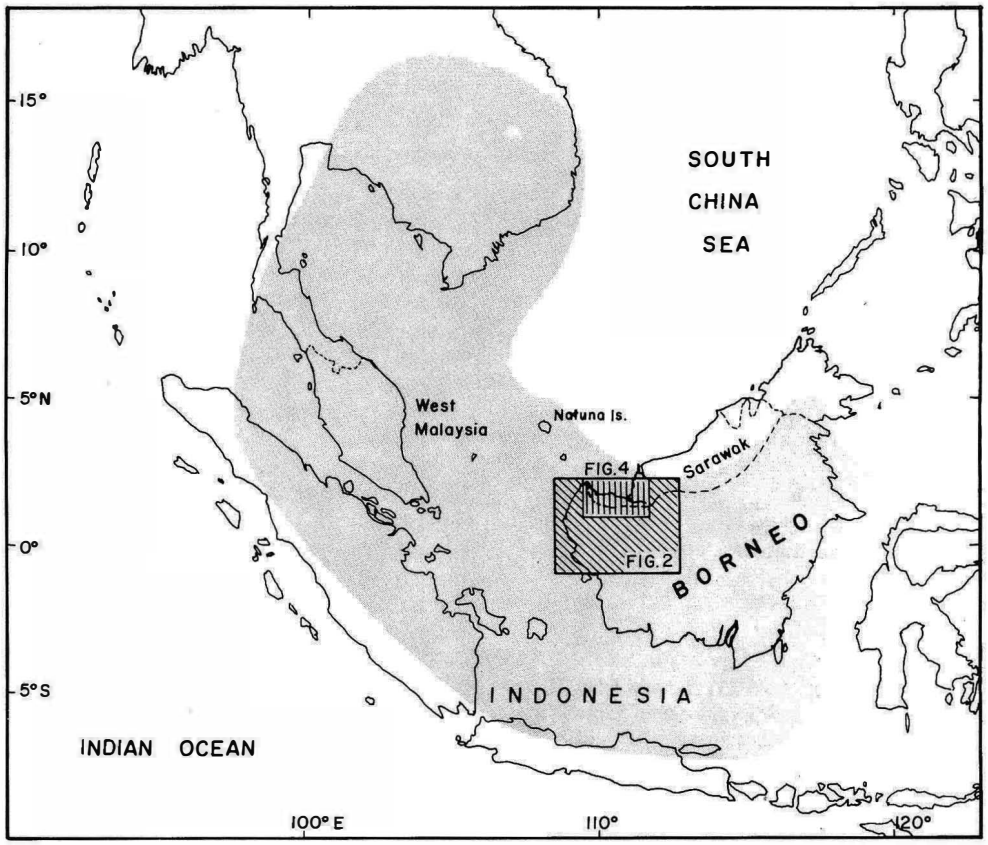


Fig. 1. Map of the Southeast Asian region, showing the approximate extent of Sundaland (shaded), the location of the regional geologic map in Figure 2, and the sample locality map in Figure 4. The borders of Sundaland and the extent of its Cenozoic internal deformation are disputed.

Sundaland and the entire Southeast Asian region, so paleomagnetic data from it will provide important limiting conditions on plate reconstructions. Such data may also help to define how much of Sundaland has actually behaved as a unit. We present new Cenozoic and Mesozoic paleomagnetic data from Sarawak, Malaysian Borneo (Figure 1) and use the data to test tectonic models for the Late Mesozoic and Cenozoic tectonic evolution of Sundaland.

GEOLOGIC SETTING OF WEST BORNEO

The overall geologic data base for Borneo is weak. Mapping has been mainly of a reconnaissance nature, and both radiometric and paleontologic age control are scarce. Many of the difficulties encountered are due to poor exposure in the tropical vegetation. However, enough data exists to provide some age and structural control on paleomagnetic studies. This is especially true in West Sarawak, which is one of the best mapped areas in Borneo. To place the geology of Sarawak within a regional framework we present a brief summary of the known geology of West Borneo.

Prior to 1940, Dutch geologists mapped extensively in Kalimantan (Indonesian Borneo) [van Bemmelen, 1949]. West Sarawak was mapped by British geologists in the 1950s and 1960s, and the Malaysian Geological Survey in the 1970s. Hamilton [1979] interpreted the geology of Borneo within the plate tectonic scheme for the first time. Extensive new mapping and gravity data from West Kalimantan are presented by Williams et al. [1988], who interpret northwestern Borneo (including West Kalimantan and West Sarawak; Figure 2) to consist of three distinct geologic domains: (1) the Northwest Kalimantan domain (which includes westernmost Sarawak); (2) the Schwaner Mountains domain, and (3) the melange and associated accretionary deposits domain. Sarawak lies in both domains 1 and 3 (Figure 2). For clarity of discussion, and because their interpretation is the most recent, we follow their domain scheme.

The Northwest Kalimantan domain comprises Upper Paleozoic and younger rocks consolidated into a coherent continental crustal block [van Bemmelen, 1949; Hamilton, 1979; Williams et al., 1988] (Figures 2 and 3). The pre-Jurassic rocks are too complexly deformed to evaluate paleomagnetically, so we limit discussion to the post-Triassic geology. The Serabang Formation [Wolfenden and Haile, 1963], a highly deformed, polymict assemblage interpreted as Jurassic or Cretaceous(?) subduction melange [Hamilton, 1979] is exposed in westernmost Sarawak. Similar rocks crop out on the Natuna Islands northwest of Borneo (Figure 1)

[Wolfenden and Haile, 1963; Hamilton, 1979]. Van Bemmelen [1949] assigned both exposures to the "Natuna structural zone." Hamilton [1979] considered the rocks part of an arcuate, south dipping subduction complex, which extended northeast along offshore Vietnam and farther east in Sarawak.

Shallow marine, marginal basin sediments were deposited across the Northwest Kalimantan domain in Upper Jurassic and Cretaceous time. Near-shore facies clastics and shallow marine limestones grade laterally and upward into basin fill sediments as much as 5 km thick [Wilford and Kho, 1965; Wolfenden, 1965; Muller, 1968; Williams et al., 1988] (Figures 2 and 3). These rocks are open to tightly folded about gently to moderately plunging, approximately east-west axes. Most of this deformation took place prior to the latest Cretaceous, since the rocks are unconformably overlain by the gently deformed, fluviatile Plateau Sandstone Formation, the basal part of which contains Senonian pollen (Figures 3 and 4) [Wilford and Kho, 1965; Muller, 1968; Williams et al., 1988]. Structural and stratigraphic evidence [Wilford and Kho, 1965; Williams et al., 1988] suggests that the rocks are not imbricated into a melange, as are similar rocks farther east in Sarawak [Tan, 1979]. The contact between these rocks and the Serabang melange has not been observed, so the relationship between the two remains unclear.

In the Schwaner Mountains domain (Figure 2) an extensive, E-W trending complex of granitic plutons and basaltic to andesitic volcanic rocks of probable Cretaceous age has been identified by field mapping, geophysical and geochemical data, and K-Ar dating [van Bemmelen, 1949; Haile et al., 1977; Williams et al., 1988]. Intrusive rocks, also of probable Cretaceous age, crop out in West Sarawak and the Tambelan and Natuna Islands west of Borneo [Kirk, 1968; Haile, 1970; Haile and Bignell, 1971]. Hamilton [1979] interprets this complex as a remnant magmatic arc, associated with melange and accretionary rocks, which formed above a subduction zone along northern Borneo during the Cretaceous and Early Tertiary. He infers extension of the complex into central Borneo.

The melange and associated accretionary deposits domain lies north of the Schwaner Mountains and east of Northwest Kalimantan (Figure 2). It comprises five subparallel geologically and geophysically distinct belts. Williams et al. [1986, 1988] and Tan [1979, 1982] describe two Upper Cretaceous or Lower Tertiary subduction melanges (Boyan and Lubok Antu / Kapuas) which are flanked on either side by three strongly deformed flysch sequences ranging in age from Upper Cretaceous to Eocene. The two southern flysch sequences are imbricated and probably represent parts of an

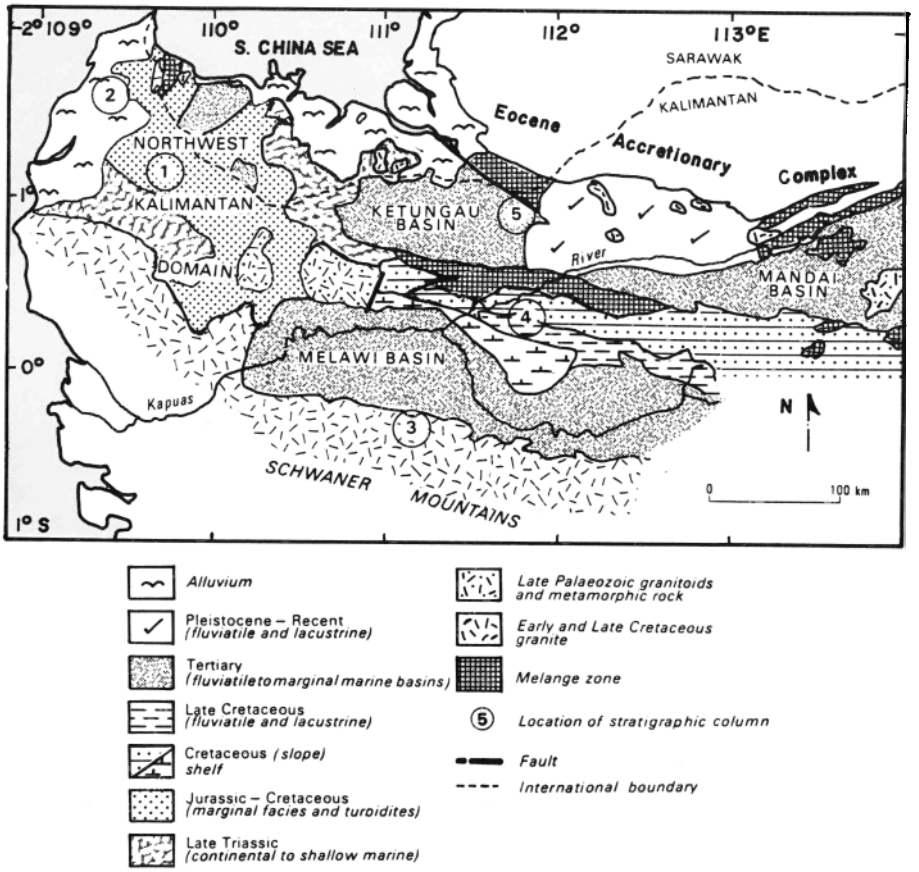


Fig.2. Geological sketch map illustrating the major geologic units and domains in northwestern Borneo. Circled numbers indicate the regions from which the generalized stratigraphic columns in Figure 3 are taken. Slightly modified from Williams et al. [1988].

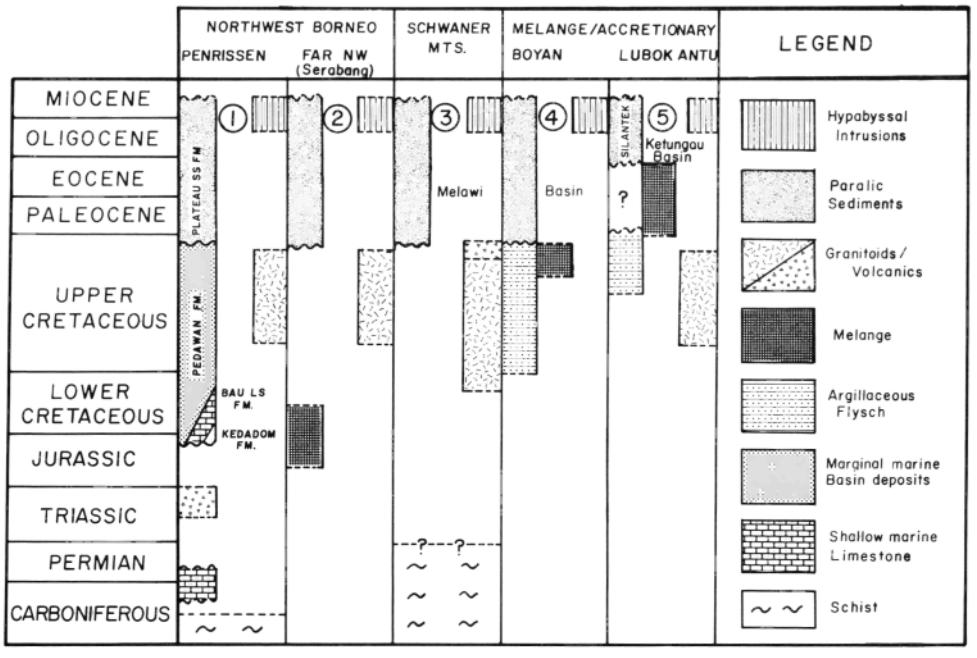


Fig. 3. Generalized stratigraphic columns from Northwest Borneo. Locations are indicated by circled numbers in Figure 2. Modified slightly from Williams et al. [1988].

accretionary prism [Williams et al., 1988]. The northernmost of the flysch sequences is very thick and forms a great crescentic outcrop across northern Borneo. Hamilton [1979] interpreted this as a massive Eocene accretionary wedge, created by southward directed subduction beneath northern Borneo (Figure 2). He describes the Eocene situation as broadly analogous to the modern setting in the Andaman-Nicobar-Sumatra Trench, where voluminous Bengal Fan sediments are being imbricated by subduction of the Indian plate beneath Sundaland.

The melange and accretionary deposits are unconformably overlain by, or faulted against, two thick, gently deformed Upper Cretaceous and Tertiary sedimentary sequences interpreted as forearc basin fill (Figures 2, 3 and 4) [Hamilton, 1979; Tan, 1979, 1982; Williams et al., 1986, 1988]. Rocks of the southerly Melawi Basin do not crop out in Sarawak. The northern basin is known as Ketungau in Sarawak and Mandai in Kalimantan. The Silantek and Plateau Sandstone Formations of West Sarawak are part of the Ketungau sequence (Figures 2, 3, and 4).

Small hypabyssal stocks (most less than 2 km in diameter), dykes and sills intrude all older rock types across western Borneo. More than 150 such bodies have been identified in Kalimantan, and many more are present in Sarawak. The rocks range from porphyritic microdiorite to microgranite and rhyolite [Kirk, 1968; Williams and Harahap, 1987]. Reported K-Ar ages, from either hornblende or biotite, vary between approximately 30 and 16 m.y. (Upper Oligocene and Lower Miocene) (Kirk [1968], Williams and Harahap [1987], and this study). The origin of the intrusions is problematic because the K-Ar ages postdate the presumed end of subduction in the area [Williams and Harahap, 1987].

PALEOMAGNETIC SAMPLING AND ANALYSIS

We present the results of 231 samples from 31 sites, collected from the Northwest Borneo Domain, the Ketungau forearc basin sediments which overly the melange and accretionary deposits, and the Tertiary hypabyssal rocks which intrude both domains (Figure 4). Samples were collected during trips in February, 1985, February, 1986 and March, 1988. Six to 10 oriented drill cores were collected at each site, and one site was collected by oriented hand specimen techniques. Samples were analyzed in the University of California Santa Barbara paleomagnetic laboratory, within a magnetically shielded room. Two instruments were used: a 2G Enterprises^R, three-axis, cryogenic SQUID magnetometer and a Molyneux Molspin^R portable spinner/fluxgate magnetometer. Samples were subjected to one of three cleaning methods. Alternating field demagnetization (AFD) was carried out using a Schonstedt^R demagnetizer capable of a peak field of 100 mT. Samples were demagnetized in steps of 5 or 10 mT until less than 1% of the natural remanent magnetization remained. This method proved particularly effective for cleaning the hypabyssal intrusive rocks.

Progressive stepwise thermal demagnetization (STD) was applied to most sedimentary rock samples. Samples were heated in a shielded furnace inside the magnetically shielded room to approximate zero field. Heating was generally done in steps of 50°C up to 300°C, then in increments of 25° to 30° until all the magnetization was removed. The third cleaning technique used was continuous thermal demagnetization (CTD) [Dunn and Fuller, 1984], which allows simultaneous heating and measurement of samples. This method was employed to

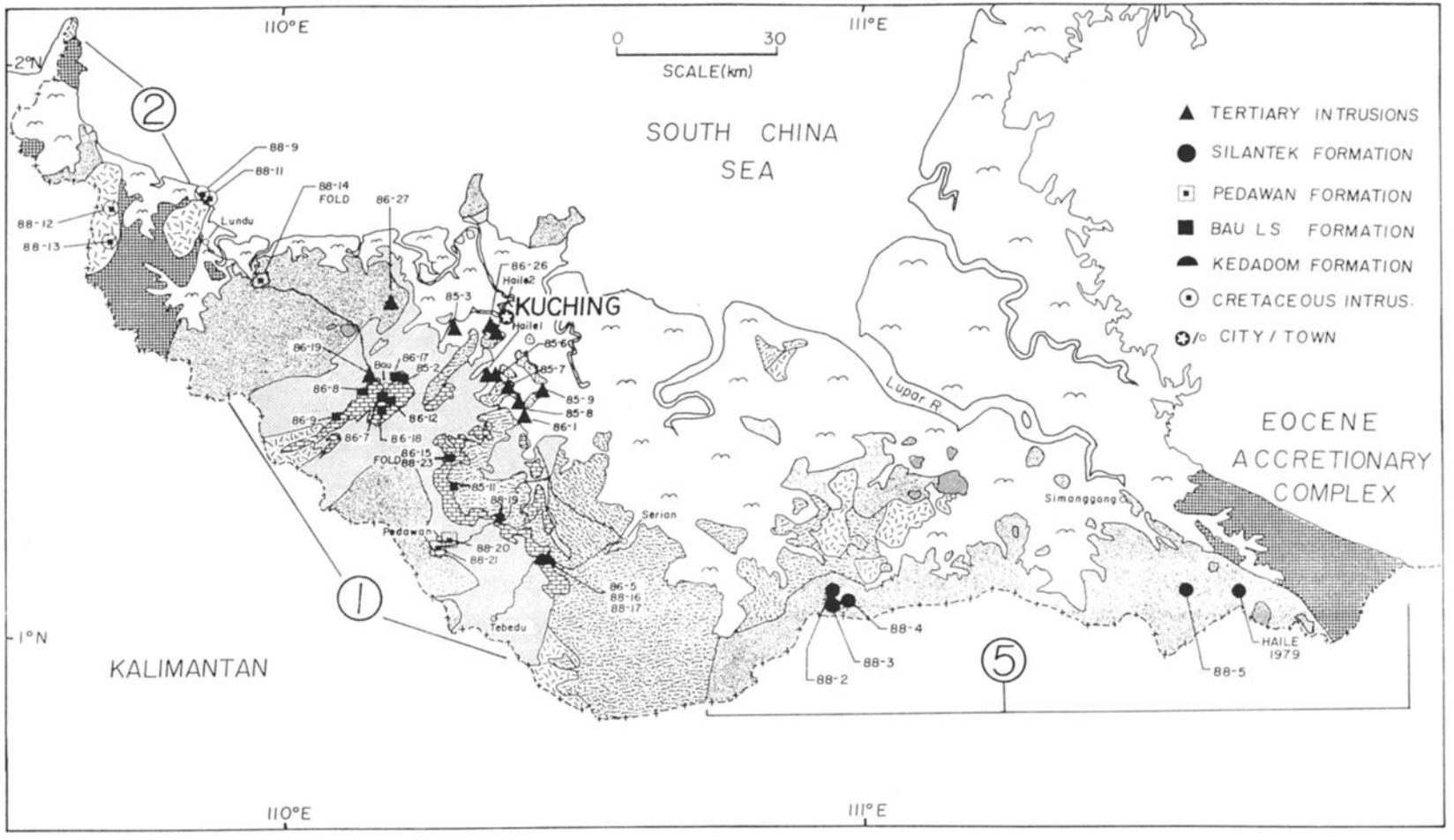


Fig. 4. Paleomagnetic sample locality map of Southwest Sarawak, with generalized geology. Sites which yielded useable results are shown, along with the unsuccessf lines refer roughly to the areas the stratigraphic columns in Figure 3. Formation names are shown on stratigraphic columns 1 and 5 in Figure 3. Geologic elements an

clean most of the rocks collected in 1988 (sites denoted SR88-).

PALEOMAGNETIC DATA FROM TERTIARY ROCKS

Tertiary Intrusions

We collected samples from 14 small stocks in West Sarawak. The locations of those which yielded useable results are shown in Figure 4. In addition, N. S. Haile (personal communication, 1986) provided us with unpublished results from two stocks. It was impossible to determine paleohorizontal from nearby strata, but of the intrusions we sampled many display vertical columnar jointing, suggesting that tilting was probably small. Nine of 14 sites have stable magnetizations (Table 1 and Figure 5). The samples yield the same characteristic stable directions using either AFD or STD. Single component magnetization is predominant, and stable end points are easily selected.

The 11 site mean directions are of both polarities. Site SR85-3 has an unexplained, anomalous direction, also found by N. S. Haile (personal communication, 1986). Four of the remaining 10 means show no declination deflection. The other six display significant counterclockwise (CCW) deflections. Five of these six sites have deflections of 8° to 26°, and the fifth is deflected 51.8° (Table 1). None of the 10 site mean declinations are significantly deflected CW. There is no apparent relationship between the degree of deflection and the either the geographic location or geologic setting of the sites (Figure 4). We obtained K-Ar (hornblende) ages for sites SR86-1 and SR86-27 (deflected 21.9° and 51.8°, respectively) which yield dates of 17.2 +/- 1.9 m. y. (Lower Miocene) and 25.8 +/- 1.9 m. y. (Upper Oligocene). These ages are concordant with those reported for the Tertiary intrusions in Sarawak [Kirk, 1968] and West Kalimantan [Williams and Harahap, 1987], which range between 30 and 16.3 m.y.

Silantek Formation

The Upper Eocene to Miocene(?) Silantek Formation is part of the Ketungau forearc basin sequence (Figures 2, 3 and 4). Although the stratigraphic succession has been established, the sparsity of diagnostic fossils results in poor age control within the formation [Haile, 1957; Tan, 1979, 1982]. We analyzed samples from five localities (Figure 4); three sites in red mudstones presumed to be high in the formation and two sites in thinlybedded, grey fluvial mudstones thought to be in the middle of it. In the four sites which carry stable remanence (Table 2 and Figure 6), CTD reveals two types of magnetic behavior. Red mudstone sites SR88-3 and SR88-4 display univectoral magnetizations carried by a mixture of magnetite and hematite. The other two sites, SR88-2 (red mudstone), and SR88-5 (fluvial mudstone), carry two-component magnetizations made up of a low-temperature (150°-450°C), less deflected component and a high-temperature (450°-630°C), more strongly deflected component. In both sites the higher temperature component is chosen as the stable direction of magnetization. Five samples from site SR88-3 are unstable, but the remaining 4 yield a mean direction close to the reverse equivalent of the local present field vector. This site lies only 1 km from site SR88-2, which shows strong CCW deflection, and both are within the upper part of the formation. The formation is not strongly deformed enough to account for the difference in deflection structurally, so the site is probably recently remagnetized. Mean directions of sites SR88-2,

TABLE 1. Paleomagnetic Data From Tertiary Intrusions, Sarawak

Site	Location	Rock Type	Lat	Long	n	Inc°	Dec°	a95°	k	K-Ar Age	VGP Lat.	VGP Long.	VGP A95°
					No Deflection								
SR85-6	J.K.R. Quarry, Kuching Airport	Microtonalite	1.42	110.33	6	2.8	2.7	5.8	176.0		87.3	200.8	5.8
SR85-9	Kuching Quarry, Mara Tuang Road	Tonalite	1.38	110.30	8	-2.1	358.9	7.2	114.0		87.3	314.7	7.2
SR86-19	Bau-Lundu Road, 3 km west of Bau	Diorite Porphyry	1.42	110.15	6	23.5	358.1	2.4	787.0		79.0	100.4	2.4
SR86-26	Jalan Rock, Kuching	Diorite Porphyry	1.50	110.32	6	15.8	185.4	14	23.9		79.0	261.0	14.0
					Deflected								
Haile1 [a]	Bukit Siabar, Kuching Area	Microtonalite	1.42	110.33	6	4.0	352.0	4.4	135.6		82.0	24.6	4.4
SR85-8	Hua Sun Quarry, Siburan	Microtonalite	1.36	110.38	10	-21.8	168.5	7.4	43.4		74.9	61.9	7.4
SR85-7 [a]	Semengo Quarry, South Kuching	Microtonalite	1.39	110.31	8	-11.8	170.7	3.6	446.0		79.7	46.8	3.6
Haile2 [a]	Mt. Choo's Garden, Kuching	??	1.39	110.31	5	19.0	154.0	13.7	31.8		61.8	356.4	13.7
SR86-1	1 km SE of Siburan, Kuap Area	Microtonalite	1.35	110.38	8	15.3	338.1	7.8	96.9	17.2+/-1.9	67.3	37.5	7.8
SR86-27	Gunung Serapi, 20 km NW of Kuching	Granodiorite	1.55	110.25	7	-0.2	128.2	3.7	269.0	25.8+/-1.9	38.2	19.2	3.7
					Anomalous								
SR85-3	Ang Chen Ho Quarry, Batu Kawa	Microtonalite	1.50	110.25	7	-53.3	308.4	4.4	190.0		30.0	339.0	4.4

Lat, Latitude (°N); Long, Longitude (°E); n, number of samples; Inc, inclination in degrees; Dec, declination in degrees; a95, circle of 95% confidence about the site mean; k, precision parameter; VGP, virtual geomagnetic pole position; A95, cirlce of 95% confidence about the VGP mean direction

[a] Collected and analyzed by N. S. Haile [Personal communication, 1986]

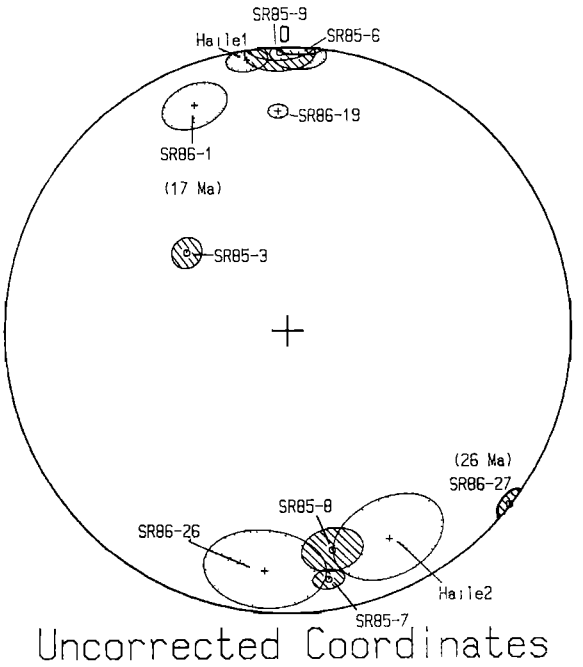


Fig. 5. Equal-area stereographic projection of the site mean directions and associated circles of 95% confidence from the Tertiary intrusions, Sarawak (Table 1). Dotted pattern with a plus in the center indicates a lower hemisphere mean; cross-hatch pattern with a circle in the center is an upper hemisphere mean. Uncorrected coordinates.

SR88-4, and SR88-5 show CCW declination deflections of 38.7° to 45.1°. The average CCW deflection is 41.4°.

Tectonic corrections for the Silantek sites are small. The site mean directions are only slightly better clustered after correction, thus it is impossible to say whether the observed magnetization is predeformational (Table 2 and Figure 6). Folding of the formation probably occurred in the Miocene, but control on the age of deformation is poor [Tan, 1979]. Whether or not the magnetizations are predeformational, the anomalous sites are in good agreement and the means display an average of 38° CCW deflection prior to tectonic correction or 41° after.

Haile [1979] reported paleomagnetic results from a site in the Silantek redbeds (Figures 4 and 6). The site mean is identical to the reverse equivalent of the present field vector. It is possible that the site comes from the youngest (Miocene?) part of the formation and is undeflected. Remagnetization due to weathering in the tropical climate is a phenomenon we have commonly observed in other redbeds from Malaysia, and it is also possible that the site was recently remagnetized.

PALEOMAGNETIC DATA FROM JURASSIC AND CRETACEOUS ROCKS

Upper Jurassic and Cretaceous Marginal Basin Deposits

The shallow marine marginal basin sequence is divided into three formations in Sarawak. The Upper Jurassic Kedadom

TABLE 2. Paleomagnetic Data From the Silantek Formation, Sarawak

Site	Location	Lat.	Long.	n	Uncorrected Coordinates				Tectonically Corrected				VGP		
					Inc°	Dec°	a95°	k	Inc°	Dec°	a95°	k	Lat°	Long°	A95°
No Deflection															
SR88-3	1 km north of Simunjam Junction	1.10	110.85	4	-14.8	190.6	6.1	229.2	-30.6	193.3	6.1	229.2	77.6	169.2	6.1
Deflected															
SR88-2	2 km north of Simunjam Junction	1.10	110.85	6	-18.9	137.4	11.8	33.4	-27	134.9	11.8	33.4	43.5	39.7	11.8
SR88-4	2 km east of Simunjam Junction	1.08	110.90	7	3.6	324.2	7	76.2	15.1	321.3	7	76.2	50.9	31.8	7.1
SR88-5	Batang Undup	1.10	111.50	9	21.3	323.2	9.6	29.9	30.8	319.6	9.6	29.9	47.3	45.1	9.6
Average of deflected sites				22	14.6	321.6	15.7	62.9	24.3	318.7	13.4	86.3	47.4	39.1	8.8

Uncorrected coordintes, site mean direction before application of tectonic correction; Corrected coordinates, site mean direction after plunge and bedding correction. Other captions are as in Table 1.

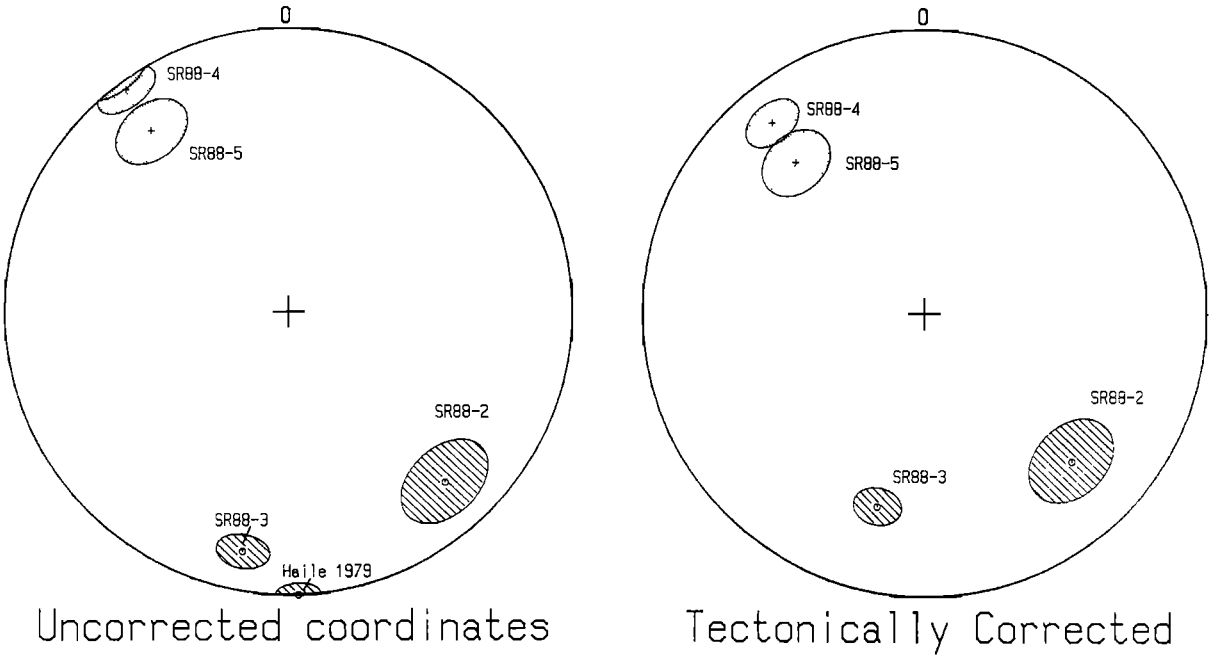


Fig. 6. Equal-area stereographic projection of the site mean directions for the Upper Eocene to Miocene(?) Silantek Formation sites, before and after tectonic correction (Table 2). Symbols are the same as Figure 5. Site mean of Haile [1979] is shown only in uncorrected coordinates.

Formation, exposed in isolated outcrops, is composed mainly of sandstone, with some interbedded conglomerate and dacitic tuff. Thin beds of shale, radiolarian chert, and marly limestone are also present [Wilford and Kho, 1965]. The Kedadom rocks represent the near-shore facies of the basin [Williams et al., 1988]. The Kedadom grades upward and laterally into the Upper Jurassic and Lower Cretaceous Bau Limestone Formation, mostly massive, pale grey or white, shallow marine limestone, with some bedded marls near the top of the formation [Wolfenden, 1965]. The Bau, in turn, grades laterally and upward into the Upper Jurassic to Upper Cretaceous Pedawan Formation, which comprises the majority of the basin fill (Figure 3). The Pedawan Formation, up to 5 km thick, is predominantly thinly bedded calcareous mudstones, shales and sandy turbidites, intercalated with lenses of *Orbitolina*-bearing limestone, radiolarian chert, and andesitic to rhyolitic volcanic rocks [Wilford and Kho, 1965]. We analyzed samples from each of these formations.

Pedawan Formation (Upper Jurassic to Upper Cretaceous). Samples were collected from two areas of Pedawan exposure. One is located about 15 km SE of Lundu along the Bau-Lundu highway (SR88-14). The site lies in an erosional window in the Plateau Sandstone Formation (Figure 4). Samples of claystone and silty mudstone were taken from both limbs of a moderately plunging, asymmetric, anticlinal chevron fold. Samples from the west limb show uniform demagnetization behavior and reveal stable vector end points when treated by CTD. The north limb sample demagnetization curves follow circular paths which intersect close to a common point. The point on each path nearest the intersection is chosen as the stable direction. Scatter between the two limbs dramatically decreases after plunge and bedding corrections, and the directions pass a fold test at the 99% confidence level (Figure 7 and Table 3) [McFadden and Jones, 1981]. This indicates that the magnetization was acquired prior to the Late Cretaceous, when the major deformation occurred [Wilford and Kho, 1965; Williams et al., 1988].

Two lenses of Lower Cretaceous *Orbitolina*-bearing limestone were sampled near Pedawan town (Figure 4). In site SR88-21 the sample directions are highly scattered, and no site mean could be determined. In site SR88-20 all samples display single-component demagnetization behavior under either STD or CTD, and stable directions are easily isolated. Both normal and reverse polarities are observed in the samples, indicating a positive "reversal test" at the site level. This is suggestive of a primary magnetization. The tectonically corrected site mean direction is identical to that obtained from the Bau-Lundu highway site but is of the opposite polarity (Table 3 and Figure 7). The two stable site means pass a "reversal test," which suggests that both are pre-Late Cretaceous directions. Both sites display approximately 90° of CCW declination deflection.

Bau Limestone Formation (Upper Jurassic and Lower Cretaceous). The Bau Limestone Formation is a suite of shallow marine carbonate rocks dominated by white or grey massive limestone, with darker grey bedded marls and calcarenites near the top of the formation. It forms a spectacular karst topography in the study area. Samples were collected from 23 sites, 11 of which carry stable remanence (Figure 4). The rest either have remanence below the resolution of the magnetometer or exhibit unstable magnetic behavior during demagnetization. The magnetization of the Bau Limestone is complex. Many entire sites or individual samples are partially or completely remagnetized, and it is often impossible to separate the components of magnetization. Two types of behavior are observed during STD and CTD. In

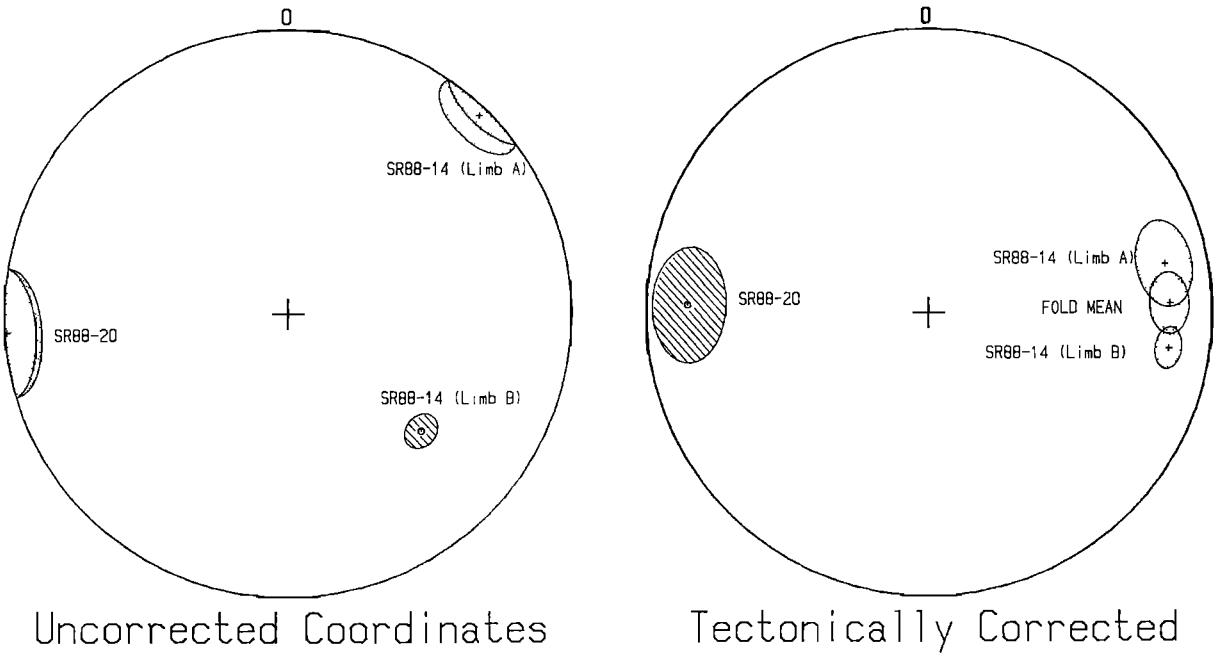


Fig. 7. Equal-area stereographic projection of the site mean directions for the Upper Jurassic and Cretaceous Pedawan Formation, before and after tectonic corrections (see Table 3). Symbols are the same as in Figure 5. The two limbs of the fold at site SR88-14 are shown separately. "FOLD MEAN" is the mean of all samples from the two limbs.

the first type, low-temperature (50°-150°C) cleaning results in removal of a soft, viscous component of magnetization, corresponding to rapid movement of the vector (as much as 60° in declination) along an approximate great circle path. The vector then stabilizes between 250° and 450°C. The second type of behavior is single-component demagnetization.

The site mean directions fall into three distinct groups (Figure 8 and Table 4). There is no clear structural or lithologic relationship between the directions and the location of the sites (Figure 4). In fact, sites with no declination deflection are often less than 1 km from those with large deflections. The first group is composed offour site means which show no significant declination deflection. Tectonic corrections do not significantly change the kappa value, thus it is unclear whether the observed magnetizations are primary or secondary. Two of the sites in this group come from recrystallized limestone within the contact aureoles of Tertiary intrusions in the Bau mining district (Figure 4). Rock magnetic experiments and thermal demagnetization indicate that the magnetization of three of the sites is carried by magnetite blocked entirely below 350°C. We believe that these three sites, and probably all four, were remagnetized between Late Miocene and Recent time by hydrothermal processes associated with emplacement of the intrusions.

The second group comprises five site means which display an average of about 60° of CCW declination deflection. Application of tectonic corrections to the group results in a large increase in scatter, indicating that this is a secondary (postdeformational) magnetization (Table 4, Figure 8). The age of the secondary magnetization is problematic. It is only possible to say that it postdates the Late Cretaceous deformation and is thus Cenozoic.

The third group in the Bau Limestone is represented by two sites, collected from both limbs of a moderately west plunging syncline near Bunuk (Figure 4). After the application of tectonic corrections the sites pass a fold test at the 95% confidence level [McFadden and Jones, 1981]. Therefore the magnetization was acquired prior to the Late Cretaceous deformation. The mean corrected direction is identical to the two directions found in the Pedawan Formation (Figure 7), also indicating 90° of CCW declination deflection.

Kedadom Formation (Upper Jurassic). Four sites were collected from the Kedadom Formation along the Serian-Tebedu road in steeply dipping, bedded, grey marly limestone (Figure 4). The sites lie on the south limb of a moderately west plunging, asymmetrical syncline [Wilford and Kho, 1965]. Three of the four sites carried stable remanence (Figure 9 and Table 3). Samples from SR86-5 were originally cleaned by STD and were later checked with CTD. Those from SR88-16 and 17 were subjected to CTD. Stable vector end points are most successfully isolated using CTD. Demagnetization behavior is very similar to that found in the Bau Limestone.

The tectonic correction is applied by simple unfolding because an accurate estimate of plunge could not be determined. The corrected average of the three means shows approximately 50° of CCW declination deflection, which is 40° less than the predeformation magnetizations in the Bau Limestone and Pedawan Formations. Because the tectonic corrections are nearly the same for all three Kedadom sites, it is impossible to tell whether we have isolated a predeformational magnetization. The lack of a plunge correction may account tor the discrepancy between the Kedadom direction and the Bau and Pedawan results.

TABLE 3. Paleomagnetic Data From the Pedawan and Kedadom Formations, Sarawak

Site	Formation / Location	Lat°	Long°	n	Uncorrected Coordinates				Tectonically Corrected				VGP		
					Inc°	Dec°	a95°	k	Inc°	Dec°	a95°	k	Lat°	Long°	A95°
	Pedawan Formation														
SR88-20	Pedawan Road, 3.5 km NW of Pedawan town	1.18	110.28	8	1.1	266.1	13.3	18.4	-15.9	271.9	13.3	18.4	1.71	12.1	13.3
SR88-14	Limb A: fold, Bau-Lundu Road, 15 km SE of Lundu	1.50	109.90	8	3.3	43.8	10	31.8	16.0	78.4	10	31.8			
SR88-14	Limb B: fold, Bau-Lundu Road, 15 km SE of Lundu	1.50	109.90	7	-37.5	131.3	4.8	160.4	15.5	98.6	4.8	160.4			
Fold Average (all samples)				15	-21	78.4	26.4	3.1	16.0	87.9	7.1	14.5	-2.3	11.8	7.1
	Kedadom Formation														
SR86-5	Serian-Tebedu Road, Gunung Merabong	1.13	110.44	8	45.3	342.8	4.4	157.0	9.7	315.5	4.4	156.7	45.4	26.3	
SR88-16	Serian-Tebedu Road, Sungai Kedadom	1.13	110.44	8	36.8	336.8	4.6	147.0	-5.3	318.2	4.6	147.0	48	15.2	
SR88-17	Serian-Tebedu Road, Sungai Kedadom	1.13	110.44	5	39.5	307.3	18.4	18.2	-10.7	299.3	18.4	18.2	29	13.6	
Average				21	41.6	328.8	23.1	2.9	-2.1	311.1	22.7	2.9	41	18.0	17.7

Captions are as in Tables 1 and 2.

OTHER PALEOMAGNETIC DATA FROM BORNEO

The few other paleomagnetic data which exist for Borneo are presented here for comparison with our results (Table 5 and Figure 10).

Tertiary. Untung et al. [1987] give paleomagnetic results from two basaltic sills of reported Oligocene or Miocene age, intruded into Mandai Basin rocks in the Nanga Raun region of Kalimantan. The mean direction is undeflected (Table 5 and Figure 10). Untung et al. also list data from sedimentary rocks, which are probably Eocene, approximately 400 km east of our study area in the Kalasin region of central Kalimantan. The mean direction shows 37° of CCW declination deflection, concordant with what we observe in the Silantek Formation (Tables 2 and 5).

Cretaceous. Haile et al. [1977] sampled volcanic and plutonic rocks in the Schwaner domain for paleomagnetism and geochronology. In most cases the rocks analyzed for paleomagnetism (granite, dikes, and hornfelses) are not dated and are assumed to be cogenetic with other dated plutons. No tectonic corrections are applied to the data, and sites are represented by single hand specimens. K-Ar ages from biotite and hornblende in the plutons yield individually concordant but widely scattered Cretaceous ages [Haile et al., 1977], while volcanic rocks in the area (some of which intrude the granite as dykes) give latest Cretaceous ages [Williams et al., 1988]. Despite large scatter, virtually all the data show CCW declination deflection. Dikes, lavas, and hornfelses show an average of 46° of deflection, plutonic rocks 56°, and all averaged 49° (Table 5 and Figure 10). In our study we analyzed four sites (SR88-9, SR88-11, SR88-12 and SR88-13) in two quartz monzonite intrusions, K-Ar dated as Late Cretaceous [Kirk, 1968]. These crop out in extreme western Sarawak (Figure 4). The rocks have extremely soft, unstable magnetizations under either AF or thermal cleaning and give scattered directions. This made determination of site directions impossible. On the basis of this information we suspect that data from the granitic rocks in the Schwaner domain may be less reliable than those from the volcanics and hornfelses.

Jurassic. Untung et al. [1987] present data from Jurassic rocks within the shallow marine marginal basin sequence in Kalimantan at latitude 0.7° N, longitude 110.1° E, about 50 km south of Sarawak (Table 5). Forty samples from 11 localities yield a mean direction which shows 93° of CCW declination deflection, indistinguishable from the predeformation directions found in the Bau Limestone and Pedawan Formations (Figure 10). This concordance indicates a pre-Late Cretaceous magnetization in the rocks.

DISCUSSION AND INTERPRETATION

Implications of the Data for Tectonic Rotation

Anomalous paleomagnetic declinations for rocks ranging in age from Upper Jurassic to Miocene from Borneo are all deflected CCW. None of the observed magnetizations show significant clockwise (CW) declination deflection, although a few show no deflection [Haile et al., 1977; Untung et al., 1987]. The data reflect a trend of increasing declination deflection with greater age (Figure 11, Table 6). Anomalous declinations in the Tertiary intrusions are deflected CCW from 9° to 52°. The most deflected intrusion was dated as 26 m. y. (SR86-27), and the one dated 17 m. y. (SR86-1) is deflected 22° (Table 1). Three sites in the Upper Eocene to Miocene(?) Silantek Formation show an average CCW deflection of 41.3°,

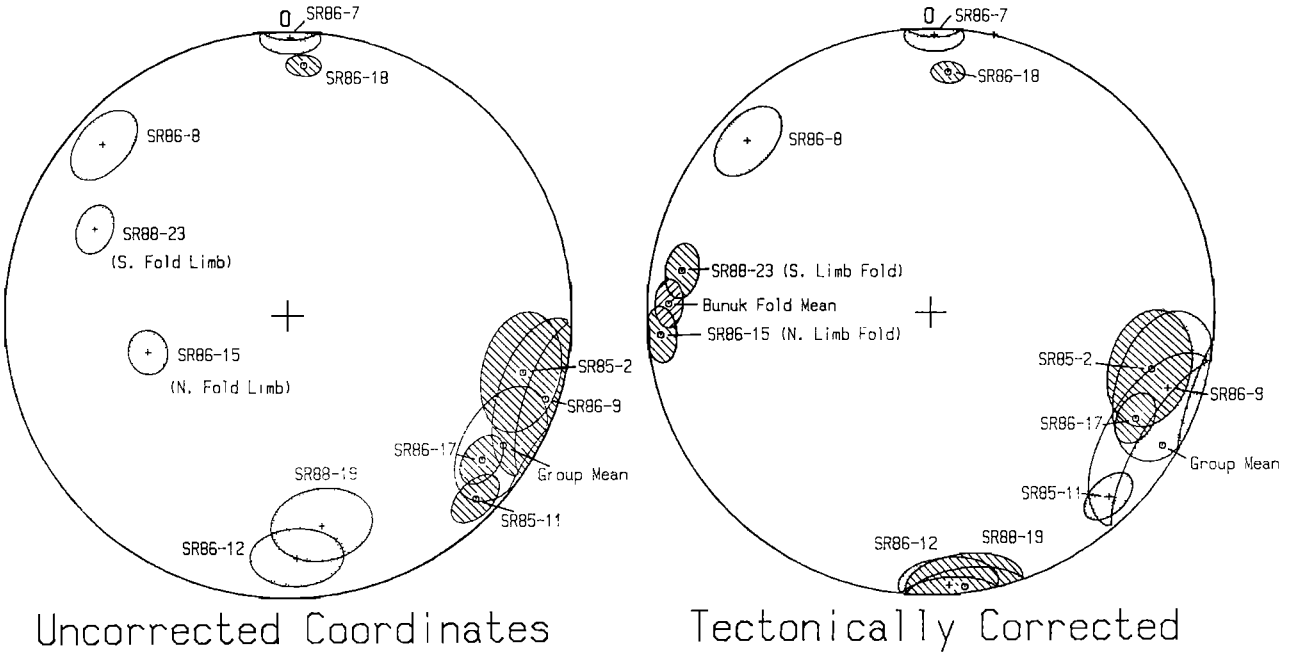


Fig. 8. Equal-area stereographic projection of the site mean directions from the Upper Jurassic and Lower Cretaceous Bau Limestone Formation, before and after tectonic correction (see Table 4). Symbols are the same as in Figure 5. The two limbs of the fold test (sites SR86-15 and SR88-23) are labeled separately. The fold mean includes all samples from the two limbs. The scatter of the Cenozoic remagnetized group increases after tectonic correction, indicating a negative regional attitude test. "Group mean" is the average of these Cenozoic remagnetized sites.

and rocks of probable Eocene age, in the Kalasin area of Kalimantan, show 37° of deflection (Tables 2 and 5) [Untung et al., 1987]. The igneous rocks of the Schwaner domain and the post-Late Cretaceous secondary direction in the Bau Limestone display similar deflections of 49° and 58°, respectively.

The declination deflections of fold-tested, pre-Late Cretaceous magnetizations in the Bau Limestone and Pedawan Formations, and Upper Jurassic rocks in the Tenguwe region of Kalimantan, are all approximately 90° CCW (Tables 3, 4, and 5). The Kedadom Formation does not fit the regional pattern in the Jurassic and Cretaceous marginal basin rocks, possibly due to an unsatisfactory tectonic correction.

The data from Borneo show consistency in the magnitude of deflection for similar age rocks over a region 400 km long east-west and at least 200 km wide north-south (Figure 11). There are several possibilities for such a trend. One is that it represents continuum deformation within a left-lateral strike-slip regime. The time period involved is probably too long to assume homogeneous stress conditions across this tectonically complex area, so this is unlikely. A second alternative is local block rotation due to simple shear in a left-lateral strike-slip system. Rotations of small (<200 km long) blocks in a broad crustal shear zone produce declination anomalies of the same sense in same age rocks but of quite variable magnitude [e.g., Ron et al., 1984; Luyendyk et al., 1985]. The magnitude of declination anomalies in the West Borneo rocks are roughly the same for same age rocks, so this explanation is implausible. Both of the forearc basin sequences, which overlie all older rock types in the area, are only gently deformed around consistent regional fold axes. Thus geologic evidence also suggests that local block rotation is unlikely in the younger rocks. We interpret the declination anomalies to represent tectonic rotation of West Borneo as a large, coherent crustal block. The data strongly suggest that West Borneo, as far as 114.2° E, and 1° S, has experienced up to 90° of CCW tectonic rotation since the Late Cretaceous. Because of the ambiguity of polarity caused by shallow inclination in equatorial rocks, it is possible that rather than CCW rotation the data imply up to 90° CW rotation. We reject this idea because of the time progression and consistent sense of declination deflection in the data.

Extent of Rotated Domain

Paleomagnetic data from Cretaceous and Lower Tertiary rocks in West Malaysia have been presented by McElhinny et al. [1974] and Haile et al. [1983] (Table 5, Figure 12). The lowermost Tertiary Segamat basalts are located in the south central part of the peninsula. They were K-Ar dated as 62 ± 2 m. y. [Bignell and Snelling, 1977] and were studied for paleomagnetism by McElhinny et al. [1974]. The 44° of CCW declination deflection they observed is consistent with that which we have found (E. A. Schmidtke et al., manuscript in preparation 1989). It must be noted that their site mean direction includes one sample from the Kuantan dykes and one from the Masai dykes, both of which they had assumed to be

TABLE 4. Paleomagnetic Data From the Bau Limestone Formation, Sarawak

Site	Location	Lat°	Long°	n	Uncorrected Coordinates				Tectonically Corrected				VGP		
					Inc°	Dec°	a95°	k	Inc°	Dec°	a95°	k	Lat°	Long°	A95°
					Late Miocene-Recent Remagnetized										
SR86-7	Mine Adit, Gunung Bau, Bau District	1.42	110.17	6	2.7	0.8	6.2	116.6	2.7	0.8	6.2	116.6	89.2	20.5	
SR86-12	North face of Gunung Mata, Bau District	1.42	110.17	9	14.9	177.8	10.5	11.9	3.7	176.1	10.5	11.9	84.9	340.2	
SR86-18	South face of Gunung Juala, Bau Distrist	1.42	110.17	6	-12.9	3.9	3.9	395.0	-16.3	4.2	3.9	395.0	79.4	87.0	
SR88-19	Kampong Tiang Bekap, Pedawan Road	1.21	110.36	10	25.4	170.8	12.3	16.5	-2.6	172.8	12.3	16.5	82.8	21.2	
Average				31	-12.7	358.6	14.6	3.9	-3.7	358.4	11.7	4.0	85.7	42.4	7
					Post-Cretaceous Remagnetized										
SR85-2	Paku Quarry, Paku Road, Bau District	1.42	110.17	7	-15.6	104.2	14.1	43.3	-20.5	104.5	14.1	43.3	14.3	28.0	
SR85-11	Kampong Giam, Kuap Area	1.32	110.28	7	-7.7	134.7	6.4	89.9	9.9	136.0	6.4	89.9	44.6	24.4	
SR86-8	Saburan Mine Entrance, Bau District	1.42	110.17	7	12.5	313.2	9.3	43.3	12.5	313.2	9.3	43.3	43.1	27.5	
SR86-9	Serikin Road, 4.6 km SW of Bidi	1.32	110.08	6	-4.8	108.5	17.5	15.7	13.1	107.8	17.5	15.7	18.5	22.2	
SR86-17	Sen Sin Ann Quarry, Paku Road, Bau District	1.42	110.17	9	-15.4	127.1	6.3	67.6	-19.3	117.5	6.3	67.6	36.9	28.9	
Average				36	-11.5	121.6	14.1	30.4	-6.1	120.0	20.9	14.4	31.5	26.1	13.7
					Pre-Late Cretaceous Magnetization										
SR86-15	North Limb Bunuk Fold, Penrissen Area	1.30	110.28	9	48.1	255	6	75.8	-5.3	265.3	6	75.8			
SR88-23	South Limb Bunuk Fold, Penrissen Area	1.30	110.28	8	26.6	294.5	6.2	80.4	-12.1	279.6	6.2	80.4			
Fold average (all samples)				17	39.7	276.6	9.6	14.8	-8.6	272.0	5.3	45.7	1.9	15.9	5.3

Captions are as in Tables 1 and 2.

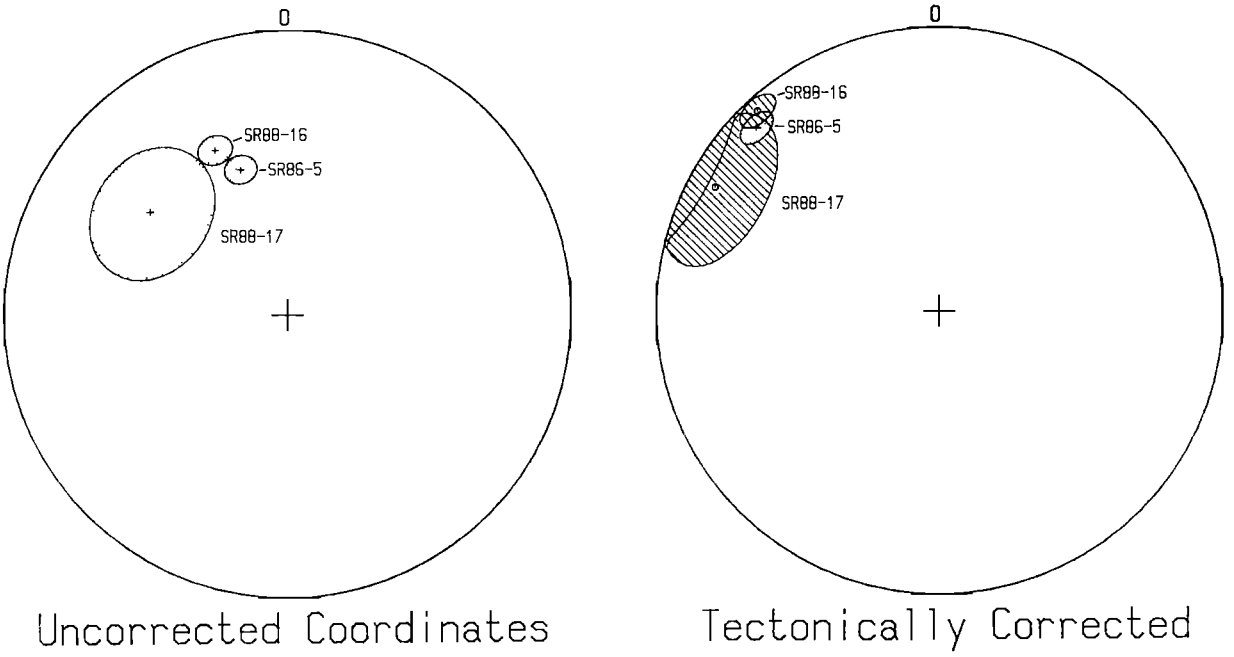


Fig. 9. Equal-area stereographic projection of the site mean directions from the Upper Jurassic Kedadom Formation, before and after tectonic correction (see Table 3). These sites were corrected for tilt only and probably have a moderate westerly plunge. Symbols are the same as in Figure 5.

TABLE 5. Other Relevant Paleomagnetic Data From Sundaland

Age	Location and Rock Type	Lat°	Long°	n	Inc	Dec	a95°	k	VGP Lat°	VGP Long°	VGP A95°
				WEST MALAYSIA							
Lower Tertiary and Cretaceous [a]	Segamat basalts [b]			16	-31.0	136.0	9.4	16.3	44.0	35.0	8.7
Cretaceous [c]	Kuantan dykes, Pahang	3.50	103.35	22	40.0	333.0	5.6	28.0	57.7	51.8	5.6
				KALIMANTAN (Indonesian Borneo)							
Oligocene or Miocene [d]	Nanga Raun (basaltic sills)	0.60	113.20	22	4.6	183.8	5.2		85.2	240.6	
Eocene [d]	Kalasin (forearc sedimentary)	0.10	114.20	27	0.6	322.8	13.9		52.8	24.5	
Cretaceous [e]	Schwaner Mts. (plutonic/volcanic)	-1.20	110.00	39	0.0	311.0	8.3	8.4	41.0	21.0	6.3
Jurassic [d]	Tenguwe (sedimentary)	0.70	110.10	40	2.6	267.1	6.9		-2.9	21.4	

Captions are as in Tables 1 and 2
[a] McElhinny et al. [1974].
[b] Includes one sample from Masai dykes, Johor, and one from Kuantan dykes, Pahang
[c] Haile et al. [1987]
[d] Haile et al. [1983]
[e] Untung et al. [1987]

the same age as the Segamat basalts. The remaining 14 samples from the Segamat basalts are in good agreement. The amount of declination deflection in Late Cretaceous [Williams et al., 1988] mafic rocks of the Schwaner domain is very similar, about 46° [Haile et al., 1977] (Table 5, Figure 11). Therefore the domain of large CCW declination deflection probably extends into West Malaysia. Assuming that the declination deflection Segamat basalts represents tectonic rotation, we suspect, as Haile et al. [1983] did, that West Borneo and West Malaysia have probably been part of a single block since the Late Cretaceous or Early Tertiary. It is possible that the agreement of the Segamat basalts with the

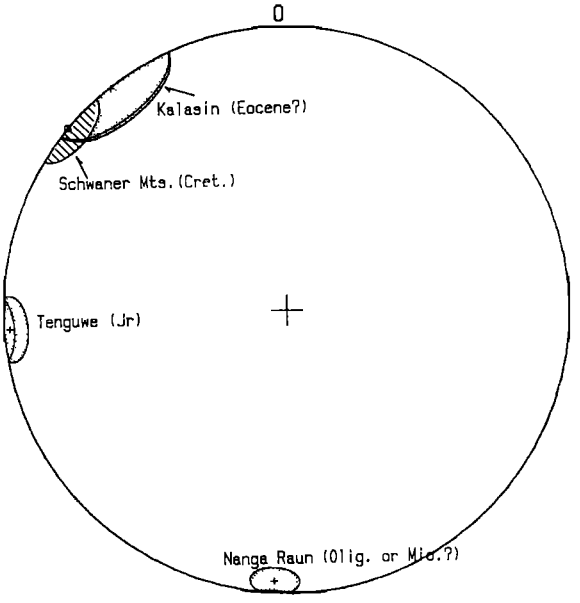


Fig. 10. Equal area stereographic projection of the mean paleomagnetic directions from Kalimantan, from the studies of Haile et al. [1977] and Untung et al. [1987] (see Table 5). Symbols are the same as in Figure 5. The data from Haile et al. [1977] did not have tectonic corrections, while the data of Untung et al. [1987] are tectonically corrected.

Borneo trend is coincidental. The lack of Cenozoic rocks in West Malaysia make additional direct tests of this idea difficult.

The Kuantan basalt dykes, located on the central east coast of the peninsula, were K-Ar dated at 104 ± 10 m. y. and studied for paleomagnetism by Haile et al. [1983]. Their mean direction displays about 30° of CCW deflection. It is clear that the Kuantan dykes do not fit the same trend as the Borneo rocks and the Segamat basalts (Figure 11). They show almost 60° less CCW rotation than the Cretaceous directions from the marginal basin sequence. This may imply that Borneo and West Malaysia have different Mesozoic histories.

Comparison With Eurasia APWP

"Stable" Eurasia, considered here includes Western Europe, Siberia, the Russian platform, South China, and North China. Kent et al. [1986] suggest that since Late Cretaceous time all the above blocks have behaved as a unit. Comparison of the Sundaland virtual geomagnetic pole positions(VGP)with the Late Mesozoic and Cenozoic synthetic APWP for "stable" Eurasia [Besse, 1986] indicates that Borneo (and possibly West Malaysia) has undergone CCW rotation with respect Eurasia throughout the Cenozoic (Table 6 and Figure 13). The translation data do not show a clear trend as the rotation data do. Only three of seven Borneo VGPs show significant translation. The largest significant translation, 12.6°, is shown by intrusion SR86-27, which has no paleohorizontal correction (Table 6). We therefore attach less importance to the translation than to the rotation data.

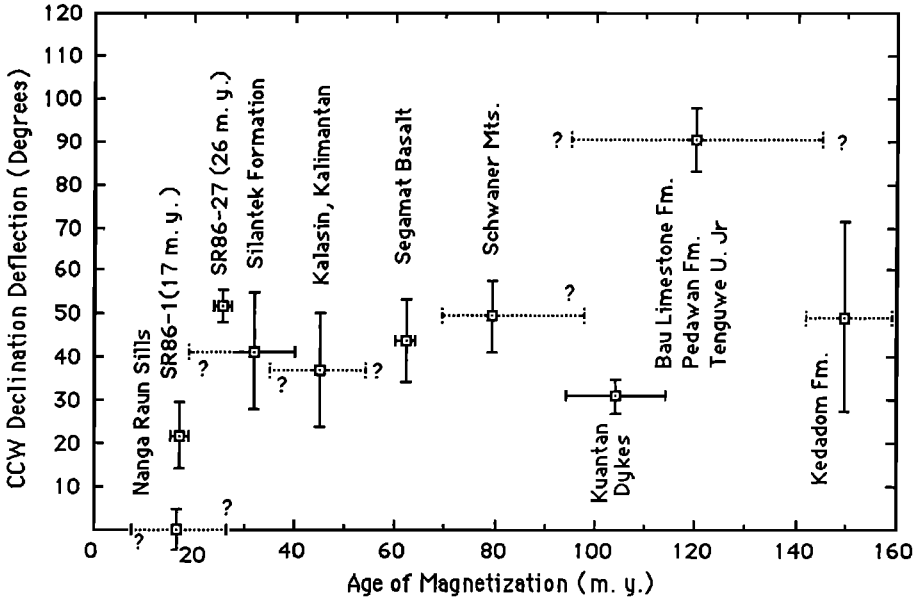


Fig. 11. Plot of CCW declination deflection versus age of magnetization for the Borneo and West Malaysia data. The magnetization in the igneous rocks (Kuantan dykes, Schwaner domain, Segamat basalts, and Tertiary intrusions) is assumed to be the same age as the rocks. The directions in the Jurassic and Cretaceous marginal basin sequence are given a minimum age by the Senonian deformation, but the magnetizations may be primary. The age of magnetization in the Silantek Formation (Table 2), Kedadom Formation (Table 3), and Kalasin Eocene rocks (Table 5) is presumed to be within the range of the age of the rocks since no stability tests exist for these data. Uncertain ages of magnetization are shown by dashed error bars.

TABLE 6. Flattening, Translation, and Rotation Calculated for Borneo Data With Respect to Stable Eurasia

Eurasia Pole	Borneo VGP	Obs Inc°	Pred Inc°	Flattening°	Trans°	Significant?	Obs Dec°	Pred Dec°	Rotation°	Significant?
20 m.y.	17 m.y. (SR86-1)	15.3 ± 7.8	20.1 ± 4.4	4.8 ± 6.5	2.6 ± 6.5	no	338.1 ± 7.8	4.4 ± 2.2	-26.2 ± 6.7	yes
30 m.y.	26 m.y. (SR86-27	0.1 ± 3.7	24.5 ± 3.9	24.2 ± 4.3	12.6 ± 4.3	yes	308.2 ± 3.7	4.1 ± 3.9	-55.9 ± 4.5	yes
40 m.y.	Silantek Formation deflected	24.5 ± 13.4	18.1 ± 3.9	-6.4 ± 11.2	-3.6 ± 7.7	no	318.7 ± 13.4	10.2 ± 3.9	-51.5 ± 12.3	yes
50 m.y.	Kalasin Eocene rocks	0.6 ± 5.2	17.2 ± 2.9	16.6 ± 6.0	8.5 ± 4.8	yes	322.8 ± 5.2	9.7 ± 2.9	-46.9 ± 4.8	yes
80 m.y.	Schwaner Domain igneous	-0.1 ± 8.3	0.3 ± 3.8	0.3 ± 9.0	0.2 ± 5.9	no	311.0 ± 8.3	14.9 ± 3.8	-63.9 ± 7.3	yes
100 m.y.	Jurassic and Cretaceous marginal basin rocks	-14.0 ± 7.3	4.0 ± 5.4	18.0 ± 10.4	9.1 ± 5.9	yes	270.6 ± 7.3	19.0 ± 5.4	-108.4 ± 7.4	yes
150 m.y.	Kedadom Formation	-2.1 ± 22.7	17.4 ± 5.7	19.5 ± 20.1	9.9 ± 14.9	no	311.1 ± 22.7	18.5 ± 5.7	-67.4 ± 18.7	yes

From synthetic Eurasian APWP [Besse, 1986]
Mean of VGPs for sites SR88-20 and SR88-14 (limestone lens and fold, Pedawan Formation) and sites SR86-15 and SR88-23 (Bunuk Fold, Bau Limestone Formation)
Obs Inc, observed inclination of site in degrees. Pred Inc, predicted inclination of site in degrees, if site were attached to stable Eurasia. Predicted from Eurasia pole.
Flattening, difference between predicted and observed inclinations. Poleward flattening is positive. Calculated from directions. Trans, poleward latitudinal translation in degrees. Poleward translation is positive. Calculated from poles. Obs Dec, declination in degrees, observed at site. Pred dec, declination in degrees, predicted from Eurasia pole. Rotation, amount of rotation between Borneo pole and Eurasia pole, in degrees. Clockwise rotation is positive. Calculated from directions. See Beck [1976] and Demarest [1983] for mathematical definition of flattening, translation,and rotation, and associated error analysis.

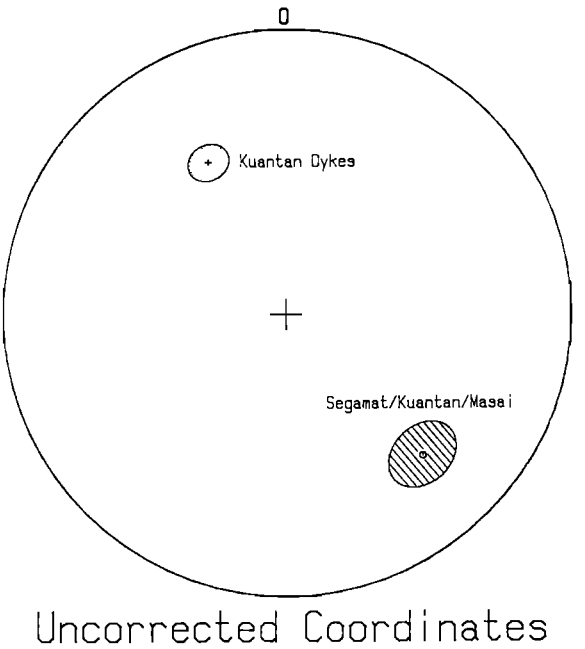


Fig. 12. Equal-area stereographic projection of the mean directions from the Segamat basalts [McElhinny et al., 1974] and the Kuantan dykes [Haile et al., 1983] (see Table 5). Uncorrected coordinates. Symbols are the same as in Figure 5.

Lin and Watts [1988] make a comparison between the Mesozoic APWPs for South China and Indochina and find just 8° of net CW rotation of Indochina with respect to the South China block since the mid-Jurassic. This means that Indochina has essentially been a part of stable Eurasia in the Cenozoic. Therefore Cenozoic relative rotation between Indochina and Borneo (and possibly West Malaysia) should be similar to that between Eurasia and Borneo.

PALEOMAGNETIC DATA AND TECTONIC MODELS OF SUNDALAND

The data presented here may be used to test proposed models for the tectonic evolution of Sundaland. Foremost among these is the "propagating extrusion tectonics" or "indentor tectonics" model of Tapponnier et al. [1982, 1986] which predicts that due to the Eocene collision of the Indian subcontinent with Eurasia, ancient sutures were reactivated as strike-slip faults, and large blocks of East Asia were extruded eastward and rotated CW. The model suggests that Indochina and Borneo may have rotated as much as 40° CW since the Eocene. The theory also relates other tectonic events, such as the 32-17 Ma opening of the South China Sea to the clockwise rotation of Indochina and Borneo with respect to Eurasia.

The data presented here do not show any Cenozoic clockwise rotation of Borneo with respect to stable Eurasia and hence do not support the "indentor tectonics" model. Tapponnier et al. [1982, 1986] argue that NW trending strike-slip faults have allowed large left-lateral displacement between South China

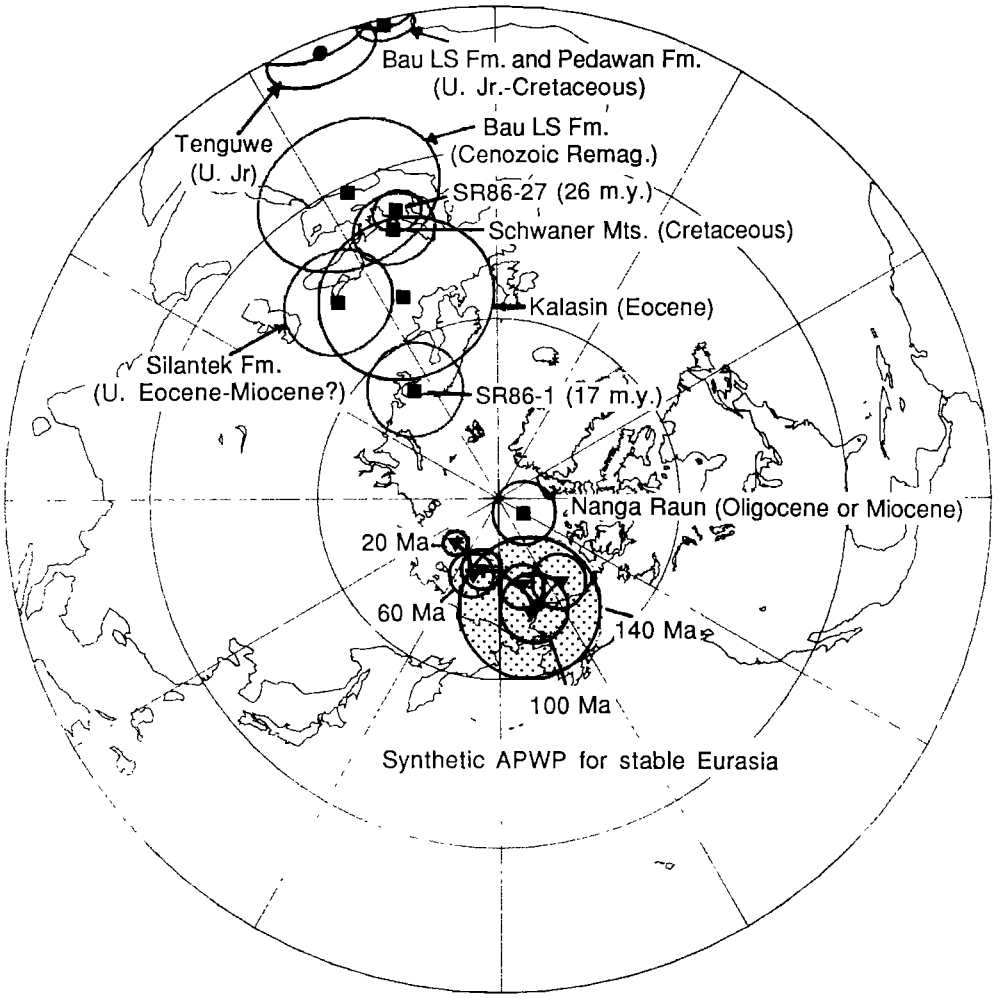


Fig. 13. Stereographic projection of the 0-140 Ma part of the synthetic APWP for stable Eurasia, shown as dotted pattern [Besse, 1986], and the VGP positions from rocks in Borneo. All the VGPs display CCW rotation with respect to the corresponding Eurasia poles (see Table 6).

and Indochina and possibly between Indochina and West Malaysia. They also argue that the same faults have probably been reactivated in a dextral sense during Neogene time. If so, it seems unlikely that the proposed 40° of CW rotation could have been completely undone by CCW rotation associated with the right-lateral motion, resulting in a net CCW rotation of Sundaland.

Although our paleomagnetic data from Sundaland show the opposite sense of rotation predicted by the indentor model, it would be naive to assume that the data imply that the collision of India with Eurasia has not had a profound effect on Cenozoic Southeast Asian tectonics. We suggest that a better model of the India-Eurasia collision can be developed if the precollision geometry of the southern margin of Eurasia is better constrained. Tapponnier et al. [1986] assume, for simplicity, that the precollision margin was a straight line. However, it is more probable that that the Early Eocene southern Eurasian margin was very irregular and that Sundaland (excluding Indochina, which, as noted above, has only rotated 8° CW with respect to Eurasia since the mid-Jurassic) extended south from Eurasia as a protruberance. It appears that a similar geometry may have existed during the Triassic, when the North China block extended south from Eurasia and the South China block collided with it (J. L. Lin and M. D. Fuller, manuscript in preparation 1989).

The second group of models are those of Hamilton [1979], Holloway [1981], Taylor and Hayes [1983], and Williams et al. [1988]. These all hypothesize Cenozoic CCW rotation of Borneo, based upon the Haile et al. [1977] paleomagnetic data, marine magnetic anomalies in the South China Sea and geologic evidence in Borneo and the South China Sea. A common feature of their theories is that as Borneo rotated, it overrode "proto-South China Sea" oceanic crust, which was subducted beneath northern Borneo. The geologic record in

West Borneo (summarized earlier) supports Hamilton's contention that south dipping subduction was fully established along northern Borneo at least as early as the Late Cretaceous. It may have continued until the Miocene, when the Palawan continental fragment arrived at the trench in northern Borneo, effectively clogging the subduction zone [Hamilton, 1979; Holloway, 1981; Taylor and Hayes, 1983]. Removing the CCW rotation of Borneo realigns the Cretaceous magmatic arc system running from South China, through Indochina, to Borneo, [Taylor and Hayes, 1983; Williams et al., 1988]. Our paleomagnetic data generally support this set of models. However, the timing and the mechanism of rotation between Indochina and Borneo (and possibly West Malaysia) are unknown. Further paleomagnetic data from Borneo, Indochina, and West Malaysia will help to clarify the extent of "stable" Sundaland and to constrain motions between the blocks.

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